



FATIN NIHAL ISLAM

EEE Graduate, KUET | AlGaIn/GaN HEMT Researcher | 2 Years Teaching Support

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Portfolio: [View Portfolio](#) **GitHub:** github.com/fatinnihal532-hub

► PROFILE

Electrical & Electronic Engineering graduate of Khulna University of Engineering & Technology (CGPA 3.88/4.00, 6th in Department, 4× Dean's Award). Undergraduate thesis on **AlGaIn/GaN HEMT device simulation in Silvaco TCAD**, supervised by Prof. Dr. Ashraful Ghani Bhuiyan, with a paper under peer review, backed by hands-on projects in power electronics, RF/antenna simulation, and electrical services design. Technical Secretary of IEEE Student Branch KUET with industrial exposure to three major power plants, including the 1320 MW Maitree STPP. Two years of teaching support experience in physics and mathematics for HSC and admission candidates. Seeking a Lecturer position in EEE, with plans to pursue graduate studies (MS/PhD) in Semiconductor Devices/VLSI.

► EDUCATION

B.Sc. in Electrical & Electronic Engineering

Session 2020-21 · Mar 2022 – Aug 2026

Khulna University of Engineering & Technology (KUET)

- CGPA 3.88 / 4.00 (6th in Department)

Higher Secondary Certificate (HSC) in Science — GPA 5.00/5.00

2018 – 2020

Saint Joseph Higher Secondary School

Secondary School Certificate (SSC) in Science — GPA 5.00/5.00

2013 – 2018

NKM High School and Homes, Narsingdi

► TEACHING EXPERIENCE

QnA Doubt Solver

Sep 2022 – Aug 2024

Udvash, Khulna

- Solved conceptual and problem-based queries in physics and mathematics for HSC and university-admission candidates.
- Kept up a high response rate and answer accuracy over two years, alongside full-time study.

► RESEARCH EXPERIENCE

Undergraduate Thesis

2025 – 2026

"Influence of Gate Architecture and Gate Position on the Performance of AlGaIn/GaN High-Electron-Mobility Transistors"

Under the supervision of **Dr. Ashraful Ghani Bhuiyan**, Professor, Dept. of EEE, KUET

- Simulated **AlGaIn/GaN HEMT** device structures in Silvaco TCAD, comparing gate geometries on on/off ratio and transconductance.
- Performed literature review, modelling/simulation, and results analysis with regular supervisor consultation.
- A paper based on this work has been submitted and is under peer review.

► LEADERSHIP EXPERIENCE

Technical Secretary

Oct 2025 – Aug 2026

IEEE Student Branch KUET

- Coordinated technical events, workshops, and seminars for the IEEE student community at KUET.
- Supported student-led knowledge-sharing sessions within the department.

Public Relations Secretary

Sep 2025 – Aug 2026

KUET TechForge

- Managed the publication activities of the departmental engineering club.
- Wrote and edited technical content covering the club's activities and achievements.

Vice-President

2025 – 2026

KUET Math Club

- Helped the committee plan mathematics-focused events for KUET students.

Joint Secretary

2025 – 2026

Dream – Voluntary Blood Donation Society of KUET

- Helped run blood donation drives and coordinate volunteers as part of the society's executive committee.

Class Representative (CR)

2024 – 2026

Department of EEE, KUET

- Elected representative of a class of 120 students — liaison between students and faculty for academic scheduling, examinations, and administrative coordination.

► ACADEMIC PROJECTS

Fire Alarm & Protection System Design — Dr. M.A. Rashid Hall

EE 4200 · 2026

- Designed a **BNBC/NFPA-compliant** zonal fire alarm and suppression system for two floors of a residential hall: 58 smoke detectors, manual call points, strobe sirens, 8-zone FACP, 156-head wet-pipe sprinkler network, FRLS cabling — **full AutoCAD layouts** and ≈ BDT 5.3 lakh cost estimate.

Microstrip Patch Antenna Design & Simulation (2.83 GHz)

EE 4206 · 2026

- Designed a circularly polarized patch antenna in **Ansys HFSS 2024 R1** — **13.3 dBi gain**, optimized 41.90 × 34.92 mm patch; parametric analysis of VSWR, radiation patterns, and impedance matching.

Single-Axis Solar Tracking System with H-Bridge Driver

EE 4110 · 2025

- Built a closed-loop tracker: **IRF540 MOSFET H-bridge** with TLP250 opto-isolated gate drive and Arduino proportional control (dead-band, sensor filtering); resolved shoot-through, gate-referencing, and transient-protection challenges.

Single-Phase Squirrel-Cage Induction Motor Design (900 W)

EE 3200 · 2025

- Complete design of a **900 W, 220 V, 50 Hz** single-phase squirrel-cage induction motor at **1500 synchronous rpm**, meeting a full-load efficiency of 0.9 and power factor of 0.82, with Class F stator winding insulation.

Single-Phase Shell-Type Transformer Design (Crossover Winding)

EE 3108 · 2024

- Complete electrical & magnetic design: core sizing, turns/conductor calculations, window-area optimization, insulation design, and loss/efficiency/regulation estimation against standard limits.

► INDEPENDENT PROJECTS (GITHUB)

AlGaIn/GaN HEMT Design-Space Explorer

Python · 2026

- Analytical HEMT model (polarization charge control, velocity saturation, access resistance) swept over **432 designs**; the on-resistance/breakdown Pareto front recovers the power-HEMT design rule.

RV32I RISC-V Core in Verilog

Verilog · 2026

- Single-cycle CPU for the full RV32I ISA with a custom assembler; **32 simulation tests**, Verilator lint, and Yosys synthesis to 3,918 gates and 992 flip-flops.

Three-Phase SPWM Inverter Model

Python · 2026

- Block-diagram RK4 simulation comparing SPWM, third-harmonic injection and SVPWM; measured a **15.4% DC-bus gain** against 15.47% in theory, with 11 checks against closed-form results.

► INDUSTRIAL ATTACHMENT & PLANT VISITS

- Bangladesh-India Friendship Power Company (Pvt.) Ltd.** — 1320 MW Maitree Super Thermal Power Plant, Rampal.
- Khulna 330 MW Dual Fuel Power Plant** (BPDB).
- Khulna 225 MW Combined Cycle Power Plant.**

► PUBLICATIONS

- Under review:** paper extending the undergraduate thesis to gate-length scaling in AlGaIn/GaN HEMTs, under the supervision of Dr. Ashraful Ghani Bhuiyan, Dept. of EEE, KUET.

► LANGUAGES

Bengali — Native · **English** — Full professional proficiency · **Hindi** — Limited working proficiency

► SKILLS

Software & Tools

MATLAB/Simulink · Silvaco TCAD · Ansys HFSS · Icarus
Verilog · Yosys · Verilator · AutoCAD Electrical · Arduino IDE
· Git/GitHub Actions · PLC Programming · MS Office

Programming

Python · C/C99 · Embedded C · C++ · Verilog HDL

Core Competencies

Circuit Design · Power Electronics · Antenna Design ·
Electrical Services & Fire Protection Design · Signal
Processing

Professional

Technical Writing · Team Leadership · Public Speaking ·
Problem Solving

Certifications

PLC Programming — Udemey

► HONORS & AWARDS

- Ranked 6th in the EEE Department, KUET (CGPA 3.88/4.00)**
- Dean's Award, KUET — Session 2023-2024 (Final Year, CGPA 3.88/4.00)
- Dean's Award, KUET — Session 2022-2023 (3rd Year)
- Dean's Award, KUET — Session 2021-2022 (GPA 3.79/4.00, 2nd Year)
- Dean's Award, KUET — Session 2020-2021 (GPA 3.92/4.00, 1st Year)
- GPA 5.00/5.00 in both SSC & HSC (Science)
- Technical Scholarship, KUET** — for academic excellence every session (2020-21, 2021-22, 2022-23, 2023-24)
- General Government Scholarship** — SSC examination
- Talentpool Government Scholarship** — PSC examination

► PERSONAL DETAILS

Present Address: 24/D, Nurul Chala, Natun Bazar, Vatara-1212, Dhaka **Nationality:** Bangladeshi

► REFERENCES

Dr. Ashraful Ghani Bhuiyan

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Undergraduate Thesis Supervisor

Dr. Naruttam Kumar Roy

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*Course Instructor — Electric Power
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► DECLARATION OF AUTHENTICITY

I do hereby declare that the information given above is true to the best of my knowledge.



Signature